

Template for Einsia Lab Paper

Navers Lab, Einsia.AI

TODO: Write the abstract.

Date: September 30, 2026

HomePage: <https://lab.einsia.ai/projectname>



1 Introduction

Video generation has advanced rapidly, with substantial improvements in visual quality, temporal coherence, and the ability to synthesize complex scenes ??????????. Beyond content creation, video models are increasingly being explored as learned world simulators that predict how environments evolve in response to actions ??????. For these simulations to support reliable planning and decision-making, the generated sequences must capture how objects move and interact under physical constraints, making physical consistency a fundamental requirement. However, current video models still produce physically implausible motions and interactions despite their strong visual realism ????. Addressing these failures requires identifying which physical phenomena models struggle to reproduce and determining whether new methods reduce these errors. Reliable physical evaluation is therefore essential for diagnosing model limitations and measuring progress toward physically consistent video generation.

Existing methods for evaluating physical consistency broadly fall into VLM-based and measurement-based approaches. VLM-based methods cover diverse phenomena, but their judgments depend on the evaluator’s own physical reasoning ??????. For example, PQSG reports that GPT-5.5 achieves 64.6% accuracy on physics questions in FinePhyEval, compared with 88.4% on object-level questions, showing that judging the physical behavior is a difficult task ?. Measurement-based methods provide quantitative scores, but some require real recordings, calibrated data, or simulator ground truth ??. Physics-IQ covers multiple physical domains, but it measures similarity to reference videos rather than checking physical laws directly, making its scores sensitive to reference quality and video alignment ??. Other methods, including Principia, measure object motion in generated videos and check whether it follows physical laws or matches the physical conditions specified in the prompt ??????. However, these tests mainly cover mechanics. These gaps motivate a systematic framework for determining which physical constraints are identifiable and directly testable from uncalibrated generated videos across different domains, without human or VLM judgments, reference videos, or simulator ground truth.

To address this challenge, we introduce **PhysScope**, a task-based benchmark for evaluating physical consistency across diverse physical domains. Our central idea is to pair controlled generation conditions with measurable physical criteria. PhysScope comprises [number of tasks] tasks spanning [physical domains], each specifying an initial scene, a physical process, and observable criteria for evaluating the generated video. For each task, we construct a first-frame image using [first-frame generation pipeline] and validate the initial conditions through [validation procedure]. We then pair the image with a prompt that describes the intended action or interaction while leaving its physical outcome unspecified. The target model generates the subsequent video conditioned on this image-prompt pair.

PhysScope evaluates each generated sequence using task-specific observables and physical criteria. We extract [observable quantities or events] through [measurement pipeline] and compute scores according to [physical constraints and scoring rules]. This design connects each score to an identifiable aspect of the generated process, while allowing different phenomena to use appropriate measurements. By combining diverse task construction with explicit evaluation criteria, PhysScope aims to preserve the breadth of flexible physical-commonsense assessment while providing the diagnostic specificity of targeted physical measurements.

We evaluate [model suite] across [number of tasks and domains] and validate the proposed metrics against [validation reference]. Our experiments reveal [principal empirical finding], while additional analyses investigate [key ablations or comparisons]. These results provide a detailed account of which physical behaviors current video models reproduce successfully and which remain challenging.

Our contributions are threefold:

- We introduce **PhysScope**, a benchmark of controlled video generation tasks spanning diverse physical phenomena, with explicit initial conditions and observable evaluation targets.
- We develop a task-specific evaluation framework that connects generated physical processes to measurable criteria, enabling interpretable assessment across different physical domains.
- We conduct a systematic evaluation of [model suite], validate the proposed measurements, and identify [principal finding about model capabilities or failure modes].

2 Method

We evaluate physical consistency through measurable constraints in generated videos. Each task specifies physical assumptions, a first frame I_0 , a continuation prompt p , and a set of metrics $\mathcal{J}_k$. A video generator G_θ and a task-specific extractor E_k form the pipeline

$$V = G_\theta(I_0, p; \xi), \quad \mathbf{z}_k = E_k(V), \quad (1)$$

where ξ denotes generation randomness and $\mathbf{z}_k$ contains the measured observables. Scoring uses explicit physical relationships without a human or vision-language model as the physics judge.

2.1 Task Definition and Physical Constraints

Generated videos generally lack absolute spatial and temporal calibration. We therefore favor constraints invariant to positive rescaling of length and time. For positive observables $\mathbf{z} \in \mathbb{R}_{>0}^m$ with length–time exponents d_{Li}, d_{Ti} , changing units gives

$$z'_i = s^{d_{Li}} \tau^{d_{Ti}} z_i, \quad D = \begin{bmatrix} d_{L1} & \cdots & d_{Lm} \\ d_{T1} & \cdots & d_{Tm} \end{bmatrix}, \quad s, \tau > 0. \quad (2)$$

Scale cancellation is expressed by the null space of D :

$$D\mathbf{a} = \mathbf{0} \implies \pi_{\mathbf{a}}(\mathbf{z}') = \pi_{\mathbf{a}}(\mathbf{z}), \quad \pi_{\mathbf{a}}(\mathbf{z}) = \prod_{i=1}^m z_i^{a_i}. \quad (3)$$

This provides a common description of length ratios, period ratios, and coupled motion–geometry tests. For example, pure rolling and small-angle pendula under the same gravity yield zero-target residuals

$$r_{\text{roll}} = \frac{v}{\omega R} - 1, \quad r_{\text{pend}} = \frac{T_1^2 L_2}{T_2^2 L_1} - 1, \quad (4)$$

where v, ω, R are translational speed, angular speed, and radius, and T_i, L_i are periods and lengths. These ratios can be evaluated directly in pixel and frame units. We also use angle, order, sign, symmetry, and projective constraints when a scalar ratio is insufficient. Tasks span mechanics, optics, electromagnetism, fluids, and thermal processes; each tests an observable consequence under specified materials, initial conditions, and geometry. Scale invariance does not remove arbitrary perspective distortion, camera motion, or nonuniform time warping.

2.2 First-Frame and Prompt Construction

We construct first frames through image generation and procedural rendering. Each scene uses a controlled view, clear boundaries, and sufficient space for the target event; colors or surface markers support identity

and rotation measurements where needed. The first frame establishes the initial state while leaving the target event to the continuation, such as providing incident rays before outgoing paths appear. Prompts specify initiation, object identity, fixed geometry, and event completion, while constraining cuts and camera motion when they hinder measurement. Pilot generations guide revisions to ambiguous inputs and separate extractor errors from generation failures. Human review is used during this construction stage. Evaluation runs retain failed continuations as well as successful ones. Because prompts may describe the intended process, scores measure physical consistency under the supplied instructions.

2.3 Automatic Measurement and Scoring

Task-specific segmentation, tracking, geometric fitting, and event detection recover trajectories, angles, boundaries, or transition times. The evaluator retains raw measurements and diagnostic overlays for inspection. For each defined metric $j \in \mathcal{J}_k$, let $m_{k,j} \in \{0, 1\}$ indicate measurement availability. An available measurement is mapped to physical agreement by

$$q_{k,j} = \phi_{k,j}(r_{k,j}) \in [0, 1], \quad S_{k,j} = \begin{cases} \lambda + (1 - \lambda)q_{k,j}, & m_{k,j} = 1, \\ 0, & m_{k,j} = 0, \end{cases} \quad \lambda = 0.15, \quad (5)$$

where $r_{k,j}$ is a physical residual or event statistic and $\phi_{k,j}$ is task-specific. Residual-based mappings decrease with deviation; event and directional criteria follow their defined rules. Internal components are combined before the availability contribution is added. Availability is distinct from correctness: a clearly tracked but incorrect trajectory remains measurable. An absent event may be scored by an occurrence test, while quantities conditional on that event may be unavailable.

To expose the two contributions, define coverage C_k and mean agreement Q_k over available metrics:

$$C_k = \frac{\sum_{j \in \mathcal{J}_k} m_{k,j}}{|\mathcal{J}_k|}, \quad Q_k = \begin{cases} \frac{\sum_{j: m_{k,j}=1} q_{k,j}}{\sum_{j \in \mathcal{J}_k} m_{k,j}}, & C_k > 0, \\ 0, & C_k = 0. \end{cases} \quad (6)$$

The video and model scores are then

$$S_k(V) = \frac{\sum_{j \in \mathcal{J}_k} S_{k,j}}{|\mathcal{J}_k|} = C_k [\lambda + (1 - \lambda)Q_k], \quad S_{\text{model}} = \frac{100}{|\mathcal{V}|} \sum_{V \in \mathcal{V}} S_k(V). \quad (7)$$

Here $\mathcal{V}$ contains all evaluated videos; domain and difficulty scores use the corresponding subsets. Defined but unavailable metrics contribute zero and remain in the denominator; undefined metrics are excluded. The score therefore combines observability and physical agreement. It evaluates only the specified constraints, and extraction failure alone does not establish a physical violation.

3 Experiments

3.1 Settings

Video Generative Models

Human Evaluation Protocol

3.2 Main Results

Quantitative research. Following the fine-grained domain-level analysis used in prior physics-oriented video benchmarks ???, Table ?? reveals strong domain specialization across models. Seedance-2.5 achieves the highest overall score (53.54) and performs particularly well in Optics (63.15) and Fluid/Interfacial phenomena (57.72), which may be attributed to the strong visual cues provided by light paths, refraction patterns, fluid motion, and interface changes. MiniMax-H3 leads in Mechanics (60.64) and Electromagnetism (35.08), suggesting stronger modeling of object trajectories, collisions, and relational interactions. Wan2.2-I2V-A14B

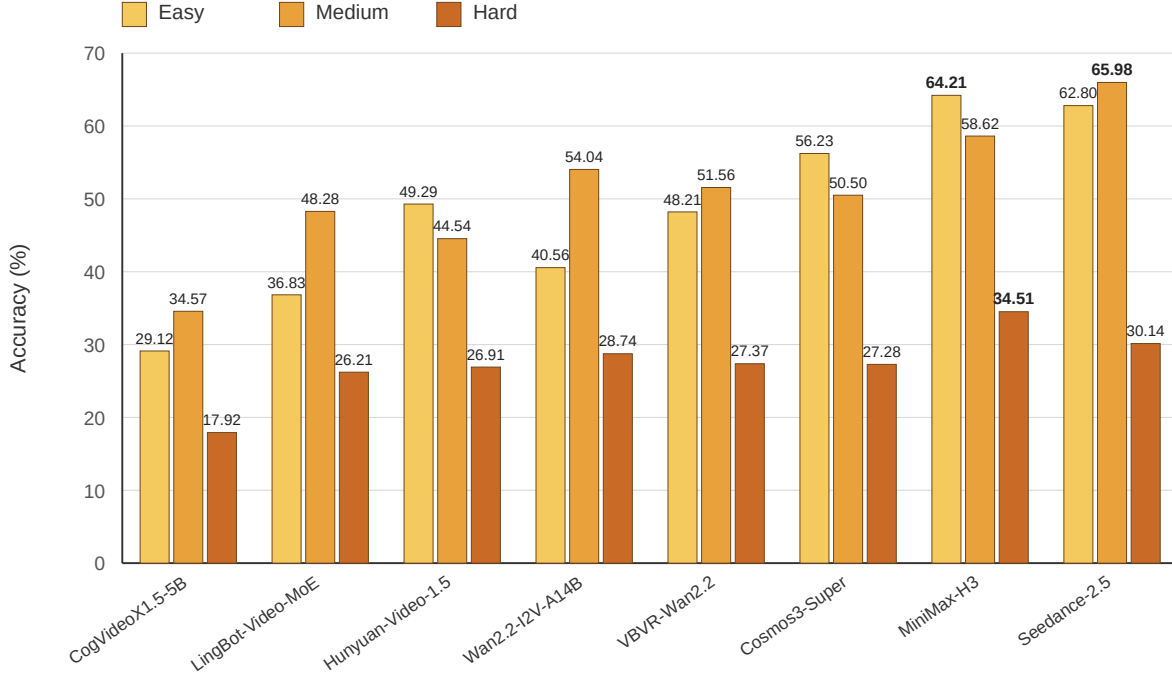


Figure 1 Performance of each video generation model across Easy, Medium, and Hard physical phenomena. The best result for each difficulty level is shown in bold.

achieves the highest Thermal score (67.91), possibly because the evaluated thermal phenomena involve visually salient state and volume changes. In contrast, Electromagnetism is the most challenging domain (24.19), likely because electromagnetic effects are often indirect and require reasoning about invisible fields, currents, and induced interactions. CogVideoX1.5-5B obtains the lowest overall score (27.79), with particularly weak performance in Optics (18.99), Electromagnetism (17.91), and Fluid/Interfacial phenomena (28.83), indicating limited capability across both visually dynamic and abstract physical processes. LingBot-Video-MoE also performs poorly in Optics (19.03) and Electromagnetism (21.98), while VBVR-Wan2.2 performs relatively well on Fluid/Interfacial phenomena but remains weak in Electromagnetism (15.03), suggesting that visually observable motion is easier to model than field-mediated physical interactions.

Figure 1 shows a clear difficulty-dependent performance pattern across all models. Easy and Medium phenomena consistently receive higher scores than Hard phenomena, because Hard cases involve longer temporal dependencies, multiple interacting objects, and stricter constraints on conservation, equilibrium, or changing motion states ????. Seedance-2.5 achieves the highest Medium score (65.98), suggesting strong performance on moderately complex and visually explicit physical processes. In contrast, MiniMax-H3 obtains the highest Easy and Hard scores (64.21 and 34.51, respectively), indicating greater robustness across both simple and highly challenging cases. The performance drop on Hard phenomena is particularly evident for lower-ranked models. CogVideoX1.5-5B achieves only 17.92 on Hard cases, which may result from the accumulation of object deformation, identity instability, and incorrect dynamics over longer or more complex sequences.

3.3 Analysis of Human Alignment

3.4 Analysis of Evaluator

3.5 Analysis of Evaluator

4 Related Works

Video Generation Evaluation. Evaluating video generation requires assessing visual quality, temporal coherence, and semantic alignment. VBench (?) organizes these aspects into multiple dimensions, including subject consistency, motion smoothness, and spatial relationships, with evaluation protocols validated against human judgments. Physics-oriented benchmarks further examine whether generated events follow physical principles. VideoPhy (?) evaluates physical commonsense in everyday activities, while PhyGenBench (?) introduces tasks spanning multiple physical domains and a hierarchical evaluation framework based on vision-language models. PhyGround (?) develops observable, law-specific criteria supported by human annotations and a specialized VLM evaluator. Complementary measurement-based approaches provide quantitative diagnostics: Principia (?) tests relational constraints between objects in Newtonian experiments, while GAUGE (?) uses measured trajectories, calibrated physical metadata, and task-specific observables to assess physical fidelity. Building on these directions, our benchmark organizes controlled experiments across mechanics, optics, and other physical domains, grounding physical scores in explicit constraints on measurable quantities. This design supports interpretable, phenomenon-specific assessment without requiring a VLM to judge physical correctness.

Physics-Aware Video Generation. Incorporating physical structure into video generation has been explored through explicit simulation and learned representation priors. PhysGen (?) combines rigid-body simulation with video diffusion, using estimated scene geometry and physical parameters to generate motion under specified forces and torques. VideoREPA (?) transfers knowledge from video foundation models through spatiotemporal token-relation alignment, improving the physical plausibility of generated videos. These approaches motivate evaluation protocols that can identify which physical relationships are preserved in generated outputs. Such assessment should cover both dynamic processes and constraints involving optical geometry, equilibrium, and conservation. Our work contributes a common testbed for examining these behaviors through controlled conditions and explicit physical measurements, enabling detailed analysis of the capabilities and limitations of video generators.

5 Conclusion

TODO: Write the Conclusion

Task			Level								
				2.5	H3	3	Wan2.2	2.2	1.5	30B-A3B	1.5-5B
1. Translational Motion and Collisions											
P1	Free fall	E		27.16 ✓	24.48 ✓	24.52 ✓	21.56 ✓	20.70 ✓	17.10 ✓	20.60 ✓	15.34 ✗
P2	Projectile motion	E		29.43 ✓	39.02 ✓	16.44 ✓	30.20 ✓	27.07 ✓	33.30 ✓	23.83 ✓	17.68 ✗
P3	Complementary-angle projection	M		53.82 ✓	58.19 ✓	21.67 ✓	15.00 ✓	15.00 ✓	15.50 ✓	38.39 ✓	13.50 ✗
P4	Bounce-height decay	M		79.32 ✓	83.21 ✓	56.23 ✓	15.00 ✓	78.76 ✓	73.52 ✓	54.67 ✓	23.75 ✗
P5	Equal-mass head-on collision	E		25.37 ✓	26.06 ✓	15.00 ✓	22.84 ✓	21.16 ✓	22.30 ✓	16.26 ✓	17.85 ✓
2. Rolling, Friction, and Rigid-Body Statics											
P6	Pure rolling of a solid sphere	M		70.93 ✓	51.20 ✓	29.86 ✓	31.74 ✓	35.39 ✓	41.78 ✓	24.39 ✓	19.46 ✓
P7	Solid sphere vs. hoop	H		25.57 ✓	27.82 ✓	24.86 ✓	28.85 ✓	18.33 ✓	30.40 ✓	20.70 ✓	17.17 ✗
P10	Rough-incline round trip	H		19.53 ✓	51.09 ✓	28.41 ✓	24.42 ✓	21.22 ✓	25.51 ✓	34.01 ✓	15.35 ✓
P42	Mass-independent sliding onset	E		36.25 ✓	97.88 ✓	98.90 ✓	36.04 ✓	34.74 ✓	14.44 ✓	57.50 ✓	35.69 ✓
P43	Toppling about a support edge	H		21.00 ✓	23.65 ✓	15.00 ✓	15.00 ✓	15.00 ✓	15.00 ✓	14.81 ✓	14.25 ✓
P44	Hanging-chain equilibrium	M		96.75 ✓	96.73 ✓	95.69 ✓	98.12 ✓	95.67 ✓	95.10 ✓	95.35 ✓	95.24 ✓
3. Pendulum Motion and Oscillations											
P8a	Small-angle isochronism	E		82.60 ✓	51.35 ✓	73.83 ✓	15.00 ✓	37.60 ✓	47.23 ✓	21.27 ✓	37.14 ✗
P8b	Mass-independent pendulum period	E		57.16 ✓	77.10 ✓	53.89 ✓	47.98 ✓	43.61 ✓	62.81 ✓	20.73 ✓	17.85 ✗
P8c	Large-angle pendulum period	H		57.57 ✓	79.88 ✓	62.81 ✓	58.72 ✓	53.39 ✓	73.33 ✓	46.92 ✓	34.21 ✗
P9	Period-length relationship	M		67.12 ✓	65.27 ✓	61.46 ✓	55.20 ✓	66.61 ✓	23.50 ✓	32.09 ✓	26.33 ✗
4. Optics and Projective Geometry											
P11	Light refraction	M		63.83 ✓	64.50 ✓	46.98 ✓	53.75 ✓	57.77 ✓	18.44 ✓	14.81 ✓	14.62 ✓
P12	Critical angle for total reflection	H		25.67 ✓	34.58 ✓	17.99 ✓	32.92 ✓	23.35 ✓	37.07 ✓	31.32 ✓	28.85 ✓
P13	Light reflection	E		95.96 ✓	35.91 ✓	15.00 ✓	95.76 ✓	15.00 ✓	34.93 ✓	15.00 ✓	14.62 ✓
P14	Point-source projection	M		67.14 ✓	15.00 ✓	30.02 ✓	15.00 ✓	27.96 ✓	15.00 ✓	15.00 ✓	17.86 ✓
concurrency											
P16	Collinear-point cross-ratio	M		80.91 ✓	82.97 ✓	82.01 ✓	83.74 ✓	67.93 ✓	88.15 ✓	85.95 ✓	14.62 ✓
5. Hydrostatics and Buoyancy											
P18	Liquid surface vs. gravity	M		66.80 ✓	72.68 ✓	19.57 ✓	47.64 ✓	28.39 ✓	37.91 ✓	59.77 ✓	22.02 ✓
P19	Communicating-vessel equilibrium	E		97.01 ✓	98.16 ✓	97.70 ✓	99.01 ✓	93.09 ✓	95.44 ✓	97.22 ✓	27.62 ✗
P20	Floating-ice submerged fraction	M		56.31 ✓	79.60 ✓	65.71 ✓	91.77 ✓	68.47 ✓	47.33 ✓	57.34 ✓	59.65 ✓
6. Phase Transitions and Melting											
P21	Water level during ice melting	M		95.77 ✓	49.07 ✓	56.86 ✓	66.94 ✓	56.36 ✓	44.68 ✓	44.83 ✓	35.12 ✓
P21b	Melting ice containing a stone	M		68.12 ✓	15.00 ✓	25.62 ✓	15.00 ✓	25.62 ✓	43.47 ✓	68.12 ✓	46.69 ✓
P21c	Freshwater ice melting in saltwater	M		51.48 ✓	48.31 ✓	60.35 ✓	59.67 ✓	61.19 ✓	47.51 ✓	41.08 ✓	25.25 ✓
P23	Freezing-induced expansion	M		85.07 ✓	92.23 ✓	89.00 ✓	85.65 ✓	87.38 ✓	67.63 ✗	85.65 ✓	81.43 ✓
P27	Crushed vs. intact ice melting	H		43.14 ✓	32.31 ✓	15.00 ✓	35.44 ✓	48.43 ✓	48.52 ✓	33.67 ✓	21.69 ✓
7. Electrostatics, Magnetism, and Electromagnetic Induction											
P28	Charged-sphere equilibrium	E		78.16 ✓	97.17 ✓	48.80 ✓	15.00 ✓	14.81 ✓	68.02 ✓	14.81 ✓	14.81 ✓
P34	Current-carrying wire and compass	H		15.00 ✓	15.00 ✓	14.81 ✓	15.00 ✓	15.00 ✓	14.81 ✓	14.81 ✓	14.62 ✓
P36	Eddy-current braking	M		15.00 ✓	15.59 ✓	15.00 ✓	15.20 ✓	15.98 ✓	16.43 ✓	15.35 ✓	35.67 ✓
P37	Closed vs. open jumping rings	H		23.00 ✓	20.84 ✓	14.25 ✓	15.00 ✓	36.25 ✓	15.00 ✓	57.12 ✓	14.44 ✓
P38	Solid vs. slotted plate damping	H		50.42 ✓	46.87 ✓	46.87 ✓	15.00 ✓	14.62 ✓	15.00 ✓	15.00 ✓	13.50 ✗
P39	Coil-induced light emission	H		14.81 ✓	15.00 ✓	29.17 ✓	15.00 ✓	15.00 ✓	15.00 ✓	14.81 ✓	14.44 ✓
8. Granular Media and Discharge Flow											
P40	Sandpile angle-of-repose scaling	E		98.89 ✓	95.01 ✓	97.60 ✓	98.69 ✓	97.81 ✓	97.32 ✓	81.12 ✓	92.55 ✓
P41	Sand vs. water funnel discharge	H		15.00 ✓	19.41 ✓	16.16 ✓	15.00 ✓	27.76 ✓	16.55 ✓	15.00 ✓	15.00 ✓
9. Surface Tension and Viscous Flow											
P45	Capillary rise vs. tube diameter	M		33.26 ✓	56.02 ✓	59.55 ✓	63.87 ✓	63.74 ✓	55.65 ✓	62.03 ✓	41.57 ✓
P47	Soap-bubble film curvature	H		15.00 ✓	15.00 ✓	15.00 ✓	15.00 ✓	15.00 ✓	14.81 ✓	14.81 ✓	14.81 ✓
P48	Volume-conserving droplet coalescence	M		70.11 ✓	50.96 ✓	42.96 ✓	63.18 ✓	66.45 ✓	25.60 ✓	25.97 ✓	14.81 ✓
P49	Sphere terminal speed in viscous flow	H		66.08 ✓	67.22 ✓	54.25 ✓	70.51 ✓	70.31 ✓	28.76 ✓	27.69 ✓	14.62 ✓
Average C				99.59	99.84	99.15	99.94	99.53	99.34	99.31	93.47
Average P				45.41	43.77	34.55	32.79	32.38	29.71	27.47	18.52
Average T				53.54	52.18	44.24	42.86	42.45	40.00	38.25	27.79

Table 1 Main results on VDM Bench. Overall scores T for eight models across 40 tasks, grouped by physical phenomenon; E/M/H denote Easy/Medium/Hard. Higher is better (0–100). Bold indicates the best total score within each task and the best model-average total score. Cell backgrounds interpolate from pale red (0) through pale yellow (50) to pale green (100) on a shared score scale. Within task rows, a green ✓ indicates that all available videos passed consistency ($C \geq 80$); a red ✗ indicates at least one rejection. These marks indicate temporal consistency, not physical correctness. Average rows show scores without consistency marks. Per-video totals are $T = 0.15C + 0.85P$ after a consistency pass and $T = 0.15C$ otherwise. The bottom rows give each model’s own averages: C and T are averaged over that model’s available videos (160, or 158 for Cosmos3); P excludes consistency-gated skips and retains recorded zeros from metric extraction failures. Cosmos3 has three videos on P5 and P13 instead of four; missing videos are not assigned zero scores.

6 Contributions and Acknowledgements

The contributors' names are sorted in alphabetical order of the last name.

Project Lead

Core Contributors

Contributors

Team Management

Corresponding to

Acknowledgements. This project is founded and supported by Navers Lab, Einsia.AI.

References

A Detailed Task Catalog

B Case Study

C Raw Data